Recommended reflow soldering profile

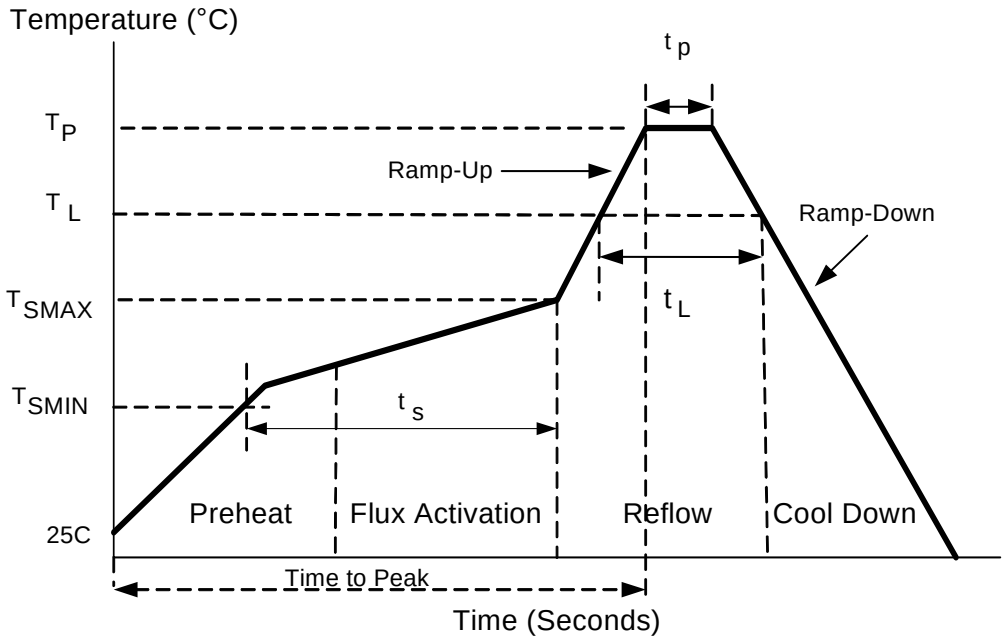

Figure 1. Temperature Profile for Infrared or Convection Reflow

Table 1. Reflow Profile

Parameter	Description	Pb-Free Package
Ramp-Up	Average Ramp-Up Rate (T_{SMAX} to T_p)	3 ºC/second max.
T_{SMIN}	Preheat Peak Min. Temperature	150 ºC
T_{SMAX}	Preheat Peak Max. Temperature	200 ºC
Tp	Max. Reflow Temperature	260 (+0/-5) ºC
T_s	Time between T_{SMIN} and T_{SMAX}	60-180 seconds
T_L	Solder Melting Point	217 ºC
t_L	Time Maintained above T_L	60-150 seconds
t_p	Time within 5 ºC of Peak Temperature	10- 40 seconds
Ramp-Down	Ramp-down Rate	6 ºC/second max.
Time to Peak	Time from 25 ºC to Peak Temperature	8 minutes max.
Reflow Cycles	Number of Allowed Reflow Cycles	3

Recommended wave soldering profile

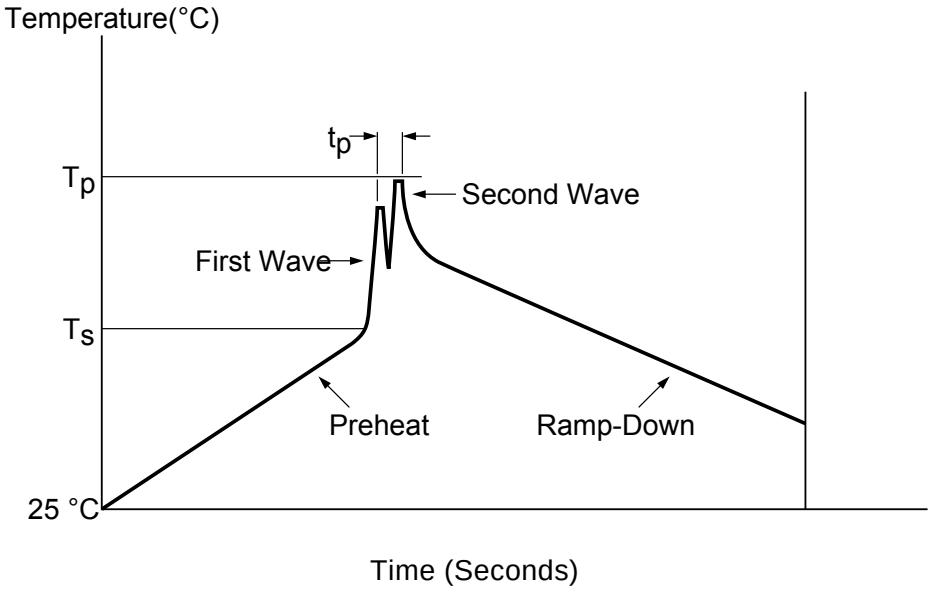

Figure 2. Temperature Profile for Wave Soldering

Table 2. Wave Soldering Profile

Parameter	Description	Pb-Free Package
Ramp-Up	Average Ramp-Up Rate	200°C/s
Heating	Heating Rate During Preheat	1 °C/s to 2 °C/s typical; 4 °C/S maximum
T_S	Final Preheat Temperature	130°C
T_P	Peak Temperature	260°C
t_P	Time Within Peak Temperature	10s
Ramp-Down	Ramp-Down Rate	5 °C/s maximum

Manual Soldering Specification

For manual soldering, MCC products can withstand 350 ºC for 3sec, pls do not exceed this limitation during soldering, otherwise the parts maybe damaged.